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Palaeomagnetism of the Askøy mafic pluton (late Precambrian), West Norway; events of Caledonian metamorphic remagnetisation

K. Rother *, P.R. Fluge and K.M. Storetvedt

Institute of Geophysics, University of Bergen, 5000 Bergen (Norway)

H. Askvik

Institute of Geology, University of Bergen, 5000 Bergen (Norway)

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Palaeomagnetic data from the mafic pluton of Askøy, West Norway, suggest that its original Precambrian magnetization(s) has been completely erased by Caledonian events of regional metamorphism. The present palaeomagnetic record defines two different axes of magnetization: a predominant component carried by magnetite (the B-remanence), with Dec = 025, Inc = −22 ($\alpha_{95} = 4°$), and a subordinate magnetization carried by haematite (the A-remanence), with Dec = 318°, Inc = 8° ($\alpha_{95} = 10°$). Comparison with Palaeozoic palaeomagnetic data for NW Europe suggests that the B-magnetization was impressed in mid-Palaeozoic time, probably in the Devonian, while the A-magnetization most likely constitutes the remains of an earlier, Upper Ordovician phase of metamorphic magnetic resetting. The palaeomagnetic evidence for metamorphic events of Ordovician and Devonian ages corroborates regional structural and reconnaissance radiometric evidence.

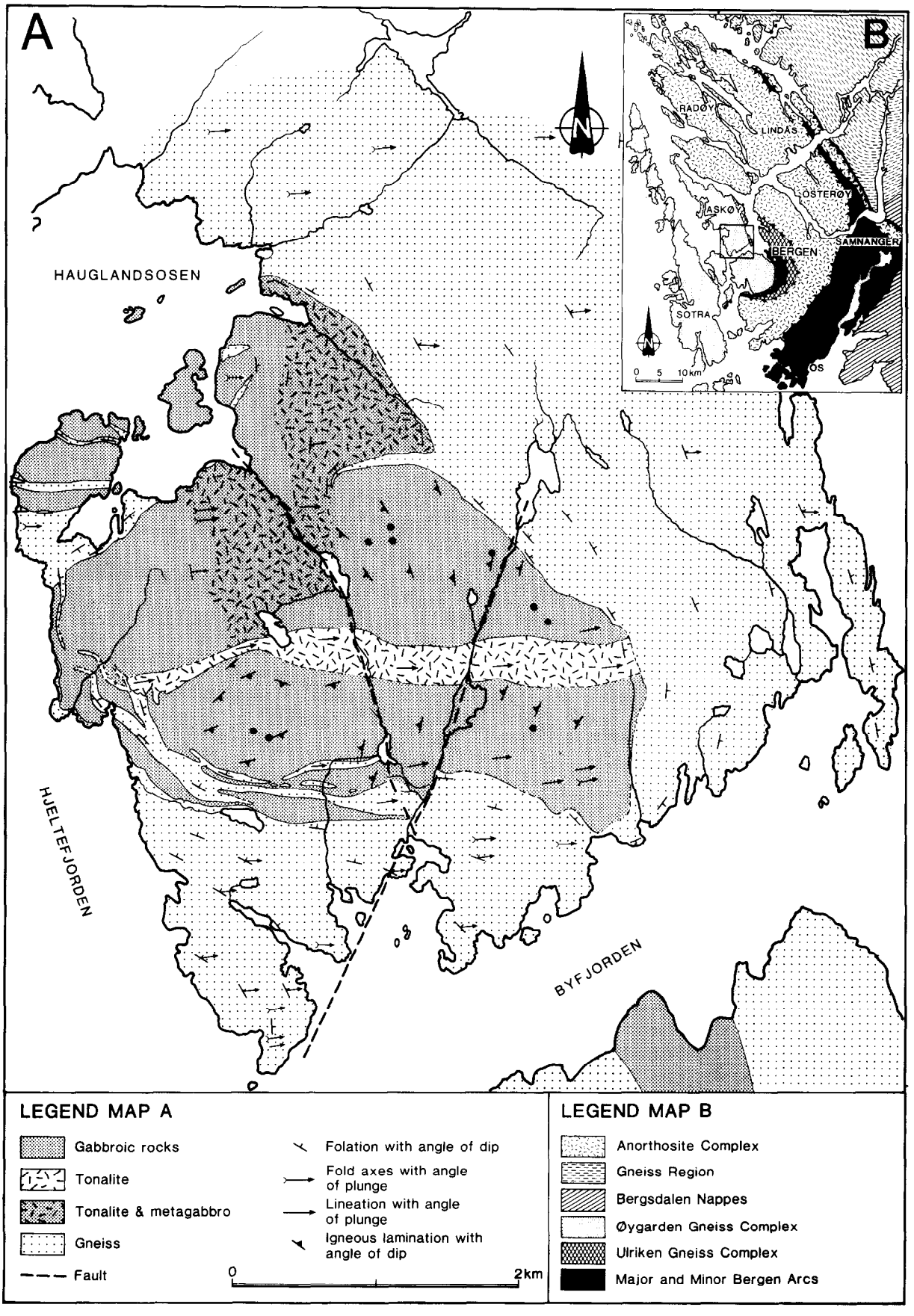
1. Introduction

The geology of the Bergen district, West Norway, comprises a series of Precambrian crystalline rocks and lower Palaeozoic schists and intrusives. East of the Øygarden Gneiss Complex (cf. Fig. 1B) the following rock units can be distinguished: the Minor Bergen Arc (metasediments and metavolcanics of assumed Cambrian–Silurian age), the Ulriken Gneiss Complex (Precambrian), the Anorthosite Complex (Precambrian), and the Major Bergen Arc (metasediments, metavolcanics etc. of lower Palaeozoic age). The tectono- stratigraphy of the area is still poorly understood, but it is now believed that the various rock units represent a pile of nappes thrust over the Precambrian basement to the east (Sturt et al., 1975; Sturt and Thon, 1978; Bryhni and Sturt, 1985). The Øygarden Gneiss Complex is surrounded by sea except for the contact to the Minor Bergen Arc, and it has not been clarified whether it is continuous to the basement to the east or represents an allochthonous unit.

Based on radiometric and structural studies various phases of deformation have been inferred (Sturt and Thon, 1978). Thus, in the Øygarden Gneiss Complex five tectonomagnetic events have been distinguished (Bering, 1985), of which the two youngest phases, D4 and D5, are of Caledonian age. D4 is thought to correspond to the concentration of radiometric ages around 450

* Present address: Central Institute of Physics of the Earth, Telegrafenberg, 1500 Potsdam, G.D.R.

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A
B
HAUGLANDSOSEN
HJELTEFJORDEN
BYFJORDEN
N
RADØY
LINDÅS
ASKØ
ØSTERØY
BERGEN
SAMNANGER
SOTRA
OS
0 5 10km
LEGEND MAP A
Gabbroic rocks
Tonalite
Tonalite & metagabbro
Gneiss
Fault
Folation with angle of dip
Fold axes with angle of plunge
Lineation with angle of plunge
Igneous lamination with angle of dip
0
2km
LEGEND MAP B
Anorthosite Complex
Gneiss Region
Bergsdalen Nappes
Øygarden Gneiss Complex
Ulriken Gneiss Complex
Major and Minor Bergen Arcs

Ma (Upper Ordovician) while D5 is a younger Caledonian event.

A fundamental problem in the interpretation of Caledonian radiometric dates from Precambrian rocks is whether the age figures represent true ages corresponding to actual orogenic events (i.e., complete resetting of the radiometric clocks), or whether we are dealing with mixed ages. However, the growing number of radiometric ages in the C.450–425 Ma range from the West Norwegian Caledonides (Neumann, 1960; Krogh et al., 1974; Sturt et al., 1975; Griffin and Bruckner, 1982) is in favour of a widespread resetting of radioisotopic systems during Upper Ordovician metamorphism. Evidence for late Caledonian orogenic processes is provided by the fact that sediments of Llandovery (Lower Silurian) age in the Major Bergen Arc have been folded under greenschist facies conditions. Furthermore, the Middle Devonian Old Red Sandstone north of Bergen has been subjected to deformation under low grade metamorphism. It is reasonable to assume that this late Caledonian event had some younging effect on the Ordovician ages.

The mafic pluton of S. Askøy intrudes the Øygarden Gneiss Complex and both share later phases of deformation. Conventional K/Ar and $^{40}Ar/^{39}Ar$ age determinations from the least altered parts of the pluton (FM Consultants, Report No. 712, 1970) gave the following results. (1) Whole-rock conventional K–Ar ages: 513 ± 31, 546 ± 22, 545 ± 22 Ma. (2) Biotite $^{40}Ar/^{39}Ar$ ages: 474 ± 14, 444 ± 13, 569 ± 17, 444 ± 13 Ma. The dating laboratory concludes that the rock is of Precambrian age, having been involved in a Caledonian metamorphic event at around 450 Ma. A Precambrian age of the plutonic mass has subsequently been given further substance by a whole-rock Rb–Sr isochron age of 890 ± 150 Ma determined from a granitic dyke intrusion that cuts the southern 'extension' of the Askøy pluton on the mainland (Sturt et al., 1975).

The problem of dating individual events in a system of superimposed metamorphic phases calls for additional information from sources other than radiometric age determination. In this respect the Askøy pluton, with its highly variable degree of penetrative deformation and mineral alteration, was considered as a potentially interesting rock formation for palaeomagnetic studies which under certain circumstances might provide essential independent information on the issue of metamorphic reworking.

2. Local geology

The Askøy pluton is a medium-grained mafic rock of gabbro–norite composition (hereafter termed gabbroic), forming a concordant, thick lens-shaped body into the surrounding Øygarden Gneiss Complex (Fig. 1A), somewhat modified by later intrusion of tonalite and by folding (Askvik, 1971). On the contacts with the surrounding gneisses no chilled margins are seen, but the rock is generally amphibolitized near the contact, so primary cooling features may have been obliterated. Gravimetric measurements show that the plutonic mass extends beneath the gneisses of the southeastern part of the island, and that its maximum thickness in the exposed area is less than 1 km (Storetvedt, 1962).

Most of the gabbroic rocks are altered and metamorphosed to varying degrees, indicated by uralitization of pyroxenes, saussuritization of plagioclase, or complete recrystallization of the rock. The various types grade into each other and their distribution is rather irregular. Macroscopically, the uralitization is indicated by the appearance of green amphibole instead of pyroxene, giving the rock a distinctive green colour, but with no essential changes in texture. Saussuritization is indicated by a light greyish-green plagioclase. The ultimate recrystallization has produced an amphibolitic metagabbro, sometimes garnetiferous. These amphibolites are characterized by a preferred orientation of elongated minerals parallel to the fold axes of the surrounding gneisses, but

Fig. 1. Geological sketch map of South Askøy (A). The inserted diagram (B) shows main geological units of the Bergen district. The palaeomagnetic locations are marked by filled circles in (A).

isolated patches of amphibolite within areas of unrecrystallized rock are massive.

Mineral lamination, generally indistinct, resulting from an alignment of platy minerals, especially feldspar laths but also pyroxenes, can be seen in patches of un-recrystallized gabbroic rock. The orientation of this igneous lamination is asymmetric, suggesting a post-consolidation westward tilting of the pluton of the order of 20° (Askvik, 1971). Alternatively, the asymmetry of primary lamination may be explained by subsidence and movement of large masses of cumulate during or after crystal settling, as proposed by Wager and Brown (1968) for the Skye intrusion.

Tonalite occurs in a large east–west striking dyke and in many minor dykes in the central part of complex, and as thick, concordant layers in the metagabbro of the northern area. The tonalite is generally foliated and gneissose with preferred eastward plunging mineral elongation parallel to the regional fold axes. The tonalite contains abundant xenoliths of metagabbro which are commonly flattened and elongated parallel to the structures of the tonalite.

Microfabric analysis of quartz axes and mica poles of a surrounding augen gneiss shows a monoclinic pattern with strong orientation, the b-axis coinciding with the regional east–west fold axes. The penetrative mineral lineation and the microfabric analysis suggest that folding took place at the latest peak of metamorphism. The mineral assemblages of the gabbroic and tonalite rocks suggest amphibolite facies conditions possibly dating from the suggested c.450 Ma metamorphic event.

The present geophysical sampling of the Askøy pluton was a dual-purpose undertaking. Material from all 43 sites collected was used in a separate study of anisotropy of magnetic susceptibility (Fluge, 1986), not reported here. However, a large part of the collected material turned out to be useless for palaeomagnetic studies (see below). From only nine of the sampled locations, all of them consisting of essentially unaltered gabbroic material, have consistent patterns of remanence directions emerged. These sites, (see Fig. 1A for aerial distribution) form the basis of the present palaeomagnetic consideration.

3. Demagnetisation results: component analysis

In the laboratory, specimens in the form of cylinders 1.9 cm in both length and diameter have been cut for measurements of intensity and direction of remanent magnetization. The measuring instruments were partly a shielded version Digico spinner magnetometer and partly a Molspin magnetometer. Alternating field (AF) treatment was carried out using a two axis tumbler, and thermal demagnetization was performed with a Schonstedt TM-1 in which the field was around 10 nT.

To define remanence directions other than that of end point magnetizations vector analysis, we carried out integrating orthogonal vector plots and the vector subtraction (v.s.) technique on all specimens defining great circle trends on progressive demagnetization. In practice, it is frequently difficult to distinguish between true straight-line segments of the orthogonal vector diagrams (the basic analytical requirement for establishing single-component magnetization) and slightly curved trajectories (signifying component overlap). This problem may be overcome by comparing vector subtracted directions for corresponding ranges of demagnetization. Thus, the presence of a 'window' in the demagnetization spectrum, for which only one remanence component is being erased, would correspond to a clustered grouping of relevant difference vector directions. On the other hand, a smeared distribution on v.s. directions demonstrates component overlap. For a meaningful application of the v.s. technique it is implied that the chosen demagnetization steps are such as to make the difference vector large enough to prevent undue influence by random measurement errors.

The foregoing section outlined that the Askøy plutonic body embraces rocks of highly variable metamorphic grade. The increasing degree of textural recrystallization is also reflected magnetically by a steady fall in bulk susceptibility (see below). The most altered rocks have also weak and inconsistent remanent magnetizations, and a low signal to noise ratio was sometimes a dominant feature throughout the full range of demagnetization. Thus, due to the measuring and experimental problems characteristic directions of magnetization have only been unravelled from the nine

gabbro locations depicted by filled circles in Fig. 1A.

Apart from a commonly encountered soft magnetization, aligned roughly along the direction of the present local geomagnetic field, the Askøy pluton is characterized by two shallow inclined axes of magnetization which are clearly discriminated by direction. The most predominant one, here termed the B-magnetization, is striking NNE, while the second axis, the A-remanence, has a NW declination. Both axes are represented by normal and reverse components but the retrieved magnetization directions are dominantly of normal polarity.

Figure 2 shows a variety of experimental and analytical results. Magnetization interactions are demonstrated by various component combinations, but in cases where A and B remanences coexist in the same specimen (cf. Fig. 2a, c, g) the A-magnetization is always the most stable one.

Table I summarizes the estimated magnetization directions for individual specimens (ignoring

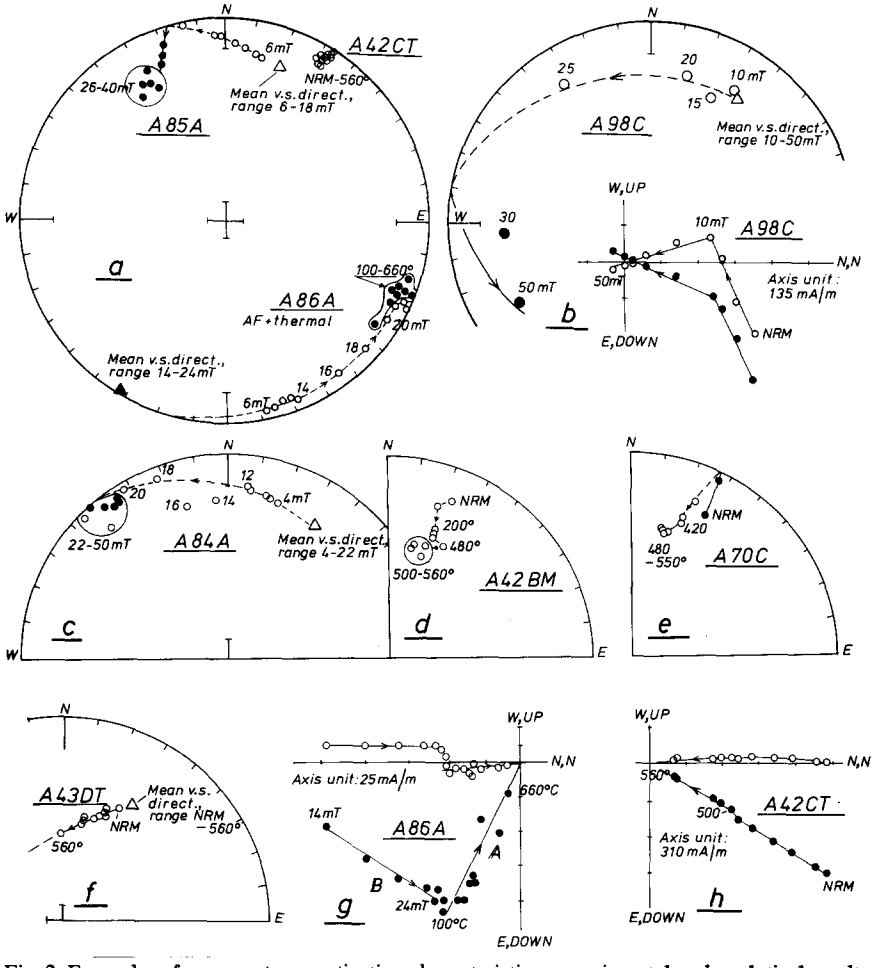


Fig. 2. Examples of remanent magnetization characteristics: experimental and analytical results. Directional plots are stereographic in which closed (open) symbols are points in the lower (upper) hemisphere. Demagnetization fields (AF) or temperatures are in milli-Tesla (mT) or in degrees Centigrade (°C). Mean vector subtracted (v.s.) directions (triangles) correspond to linear segments in orthogonal vector plots for specified demagnetization ranges. In the orthogonal vector diagrams closed (open) symbols represent vector end points in the horizontal (vertical) plane. Note that for specimen A86A (Figs. a and g) thermal demagnetization has been used on top of the alternating field treatment. Cf. text for other details.

the soft component of assumed recent origin). Figure 3 demonstrates how these results (reversed directions inverted) define the two separate A and B populations which are significantly different at the 95% significance level. The few directions that fall slightly off the encircled groups (Fig. 3) are thought to be unresolved multiphase magnetizations. The B-group which consists of high quality data is clearly much better defined than the A-group. Temperature decay patterns suggest that the B-magnetization is carried by nearly pure magnetite.

While the B-remanence is universally present in the locations concerned the A-component has only been found in three sites (the principal one being No.39). However, high temperature stable end points have been experimentally clarified for only two of the 12 A-group samples, but in both of these cases the A-magnetization can be followed to temperatures near the Curie point of haematite (i.e., A86A, Fig. 2c, g). Nine of the A-specimens show directional stability on AF treatments above c.25 mT but on further alternating field demagnetization the intensity defines a plateau which we attribute to the presence of haematite. However, subsequent thermal treatment (on top of AF) gave no further intensity reduction and only scattered results were obtained above 600 °C. The problem of establishing satisfactorily the high temperature haematite magnetization for these specimens is most likely the result of coexisting magnetite turning into a stage of magnetic instability at high temperatures, enabling the samples to pick up stray magnetization upon cooling in a small resid-

TABLE I

Magnetization directions of individual specimens (D_m, I_m) along with the corresponding palaeomagnetic group (A or B) and demagnetization range

Site	Specimen	D_m	I_m	Group	Range	Site	Specimen	D_m	I_m	Group	Range
2	A7	25	−34	B	12–18 mT	39	A83A	31	−19	B	v.s. 4–22 mT
	A8	23	−30	B	12–36 mT			354	−3	?	22–50 mT
	A9A	20	−23	B	6–33 mT		A84A	33	−15	B	v.s. 6–22 mT
	A10C	20	−34	B	3–15 mT			322	−1	A	22–50 mT
	A11A	8	−19	B	3–70 mT		A85A	20	−14	B	v.s. 6–18 mT
6	A23AT	239	11	B?	NRM–500 °C			335	−13	A	26–40 mT
	A32AT	332	−9	A	v.s.NRM–590 °C		A85B	32	−23	B	v.s. 8–20 mT
18	A42AT	17	−21	B	300–560 °C			328	16	A	24–40 mT
	A42BM	15	−29	B	500–560 °C		A86A	211	1	B	v.s. 14–24 mT
	A42BL	19	−34	B	200–560 °C			112	5	A	100–660 °C
	A42CM	33	−31	B	NRM–560 °C		A86B	184	−1	B?	v.s. 5–25 mT
	A42CT	32	−4	B	500–560 °C			111	−20	A	25–70 mT
	A43AL	34	−13	B	480–560 °C		A86K	303	9	A	590–660 °C
	A43AT	28	−27	B	NRM–560 °C		A87A	43	−27	B	v.s. 9–20 mT
	A43CL	19	−27	B	480–540 °C			329	25	A	20–70 mT
	A43DT	28	−25	B	v.s.NRM–560 °C		A87B	41	−13	B	v.s. 5–25 mT
19	A45	11	−16	B	340–530 °C			327	2	A	25–50 mT
24	A61T	26	−32	B	420–560 °C		A88A	36	−26	B	v.s. 10–25 mT
	A63T	17	−41	B	420–560 °C			323	6	A	25–40 mT
29	A70A	20	−23	B	480–550 °C		A88B	26	−25	B	v.s. 5–30 mT
	A70C	15	−26	B	480–550 °C			314	21	A	30–70 mT
	A70D	21	−25	B	480–550 °C	40	A94A	205	18	B	50–70 mT
	A71AL	56	−5	B?	550–590 °C		A94B	135	−24	A	10–70 mT
	A71CL	67	−10	B?	550–590 °C	41	A95A	15	−6	B	v.s. 5–50 mT
	A71CT	59	−12	B?	565–590 °C		A94A	32	−13	B	v.s. 10–30 mT
							A98B	31	−8	B	10–25 mT
							A98C	33	−17	B	v.s. 10–50 mT
							A99A	26	−24	B	v.s. 15–50 mT

v.s. denotes the mean specimen vector subtracted direction for range specified.

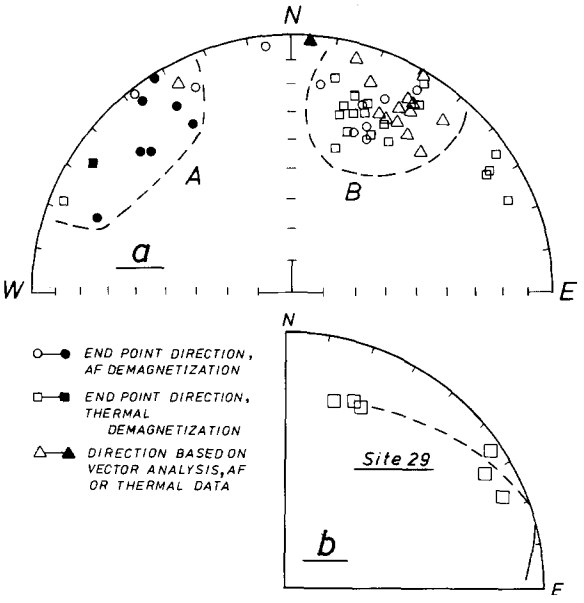


Fig. 3. Palaeomagnetic components extracted from the Askøy pluton. The encircled populations correspond to Groups A1 and B1 of Table II. The six directions of Fig. 2a, which are not included in these groups, are thought to be due to unresolved component overlap between A and B remanences. The most frequent interaction, however, is between the normal and reverse B-axis magnetizations. This probably explains the smeared site 29 distribution depicted in Fig. 3b.

ual field. Thus, the overall data quality for group A is much poorer than for group B, yet the total experimental evidence suggests that also the AF-based A directions represent relevant palaeomagnetic information.

4. Magnetomineralogy

The identity of the magnetic minerals has been inferred from thermal demagnetization data, Curie balance results, and microscopic examination of polished sections from the palaeomagnetic cores.

The microscopy data show the samples to be quite varied in terms of amount, mode of occurrence, and degree of alteration of their opaque minerals. In decreasing order of abundance these minerals are: haemoilmenite, ilmenohaematite, magnetite and iron sulphides. Figure 4 depicts photomicrographs of the most important ore phases and their textures.

'Ilmenite' which is the predominant oxide component occurs both as irregular grains and as exsolved lamellae in ilmenohaematite, sizes varying from a few millimetres down to the micron range. Ilmenite frequently shows oxyexsolution intergrowths with magnetite (Fig. 4C, D), but lamellae formation, resulting from the high temperature oxidation–exsolution process of titanomagnetite (such as are found in basalts), has not been observed. In general, the larger haemoilmenite grains consist of a dense pattern of titanhaematite lamellae (Fig. 4A, B). These titanhaematite exsolutions are particularly well developed in the interior of the original haemoilmenite, disappearing towards the surface of the grains. However, under higher magnification it is seen that the titanhaematite lamellae in turn contain exsolved ilmenite, again with the higher concentration in the interior of the parent grain. Similarly, the exsolved 'ilmenite' shows a second generation of 'haematite'. This repeated process of ilmenite/haematite exsolution suggests that the originally precipitated ilmenite phase was rich in haematite, and on subsequent cooling it became chemically unstable, leading to the formation of ferrianilmenite (with less haematite than in the parent mineral) and titanhaematite. The microscopic evidence suggests that further exsolution has finally formed nearly pure ilmenite and haematite.

Haematite has several modes of formation. Apart from the exsolution process affecting the original ferrianilmenite there are free grains of apparently homogeneous haematite, haematite formed through oxidation of magnetite (martite), and finally an important generation of haematite exsolved in the silicates of deformed gabbro.

Primary magnetite, the grain size varying upwards to the millimetre range, may occur as independent particles in the silicate groundmass but frequently it is seen to form intergrowths with ilmenite (Fig. 4C). Some magnetite grains show complete conversion to haematite, but the oxidation of magnetite is more frequently observed along cracks and grain boundaries. During penetrative deformation (a common feature of the gabbro locations of the present study) the primary magnetite has, to a varying degree, suffered textural recrystallization so that large original mag-

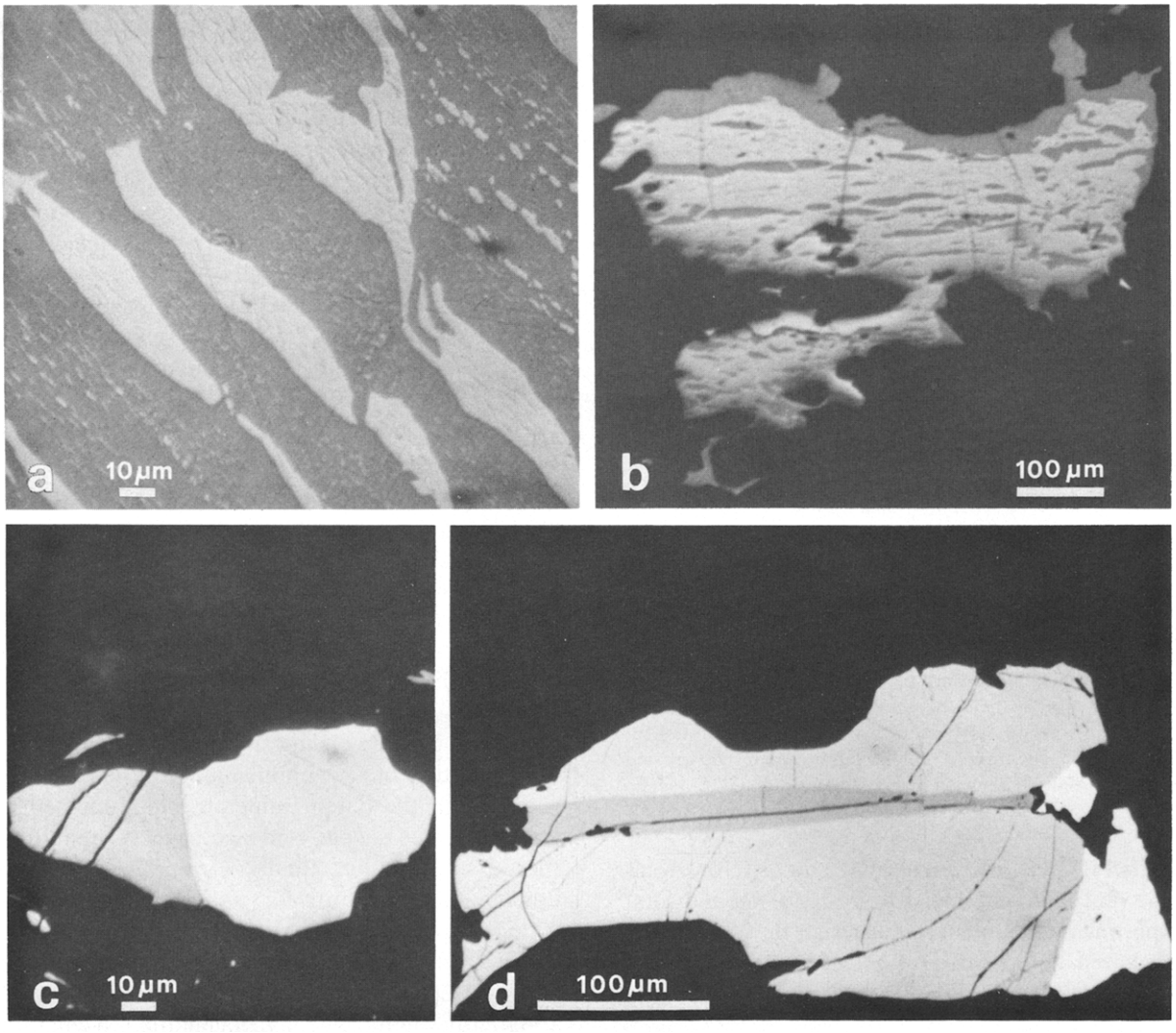


Fig. 4. Photomicrographs of opaque grains in polished sections of Askøy gabbro samples: (A) part of a large ferrianilmenite grain (dark grey) with a dense pattern of exsolved titanhaematite (light grey), the latter phase showing a new generation of small scale exsolved ilmenite; (B) grain from sample A87A (site 39) consisting of an ilmenohaematite intergrowth; (C) grain consisting of magnetite (light grey) and ilmenite (white), sample A98A (site 41); and (D) grain with magnetite (light grey), haematite (white) and a bar of ilmenite (dark grey), sample A104 (site 41).

netite particles can now be seen as a granulated system or as independent smaller grains interspersed in secondary silicate assemblages. The metamorphic disintegration process of magnetite is likely to be of prime importance for explaining the present stable magnetite remanence. Also the fine opaque dust observed in many silicate particles is probably in part magnetite.

The increasing chemical disintegration of magnetite versus increasing regional metamorphism is reflected in the bulk magnetic susceptibility in that this parameter varies from values of the order of 10^{-3} cgs units for macroscopically fresh gabbro to 10^{-5} cgs units for altered gabbro with visible secondary lineation. The optical observations suggest that rocks that have been subjected to pronounced cracking and bending, indicative of modification under shearing stress, have purer ilmenite

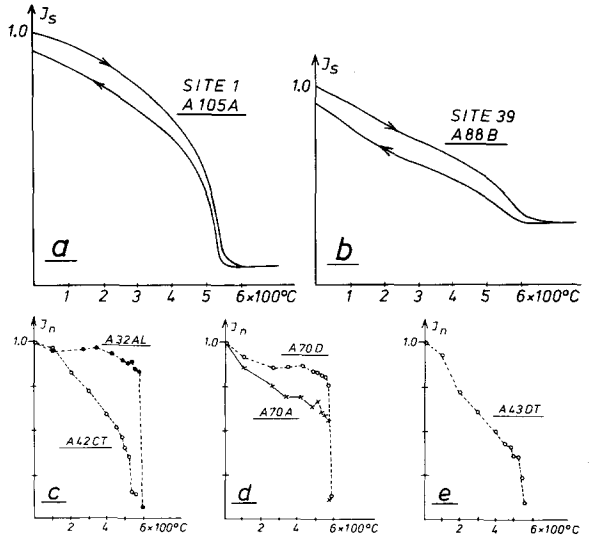


Fig. 5. Saturation magnetization (J_s) versus temperature (diagrams a and b) and examples of temperature decay of natural remanent magnetization (diagrams c–e) suggesting pure magnetite as the principal remanence carrier.

and haematite phases than those of relatively unaltered material (Fluge, 1986).

Thermal demagnetization curves, which are produced by plotting the magnetic moment measured at room temperature, after successive steps of heating and cooling in field-free space, are shown in Fig. 5c–e. Despite the fact that haematite is invariably and most abundantly present in these rocks the fossil magnetization is dominantly governed by nearly pure magnetite with unblocking temperatures in the 550–580 °C range (Fig. 5c,d). Only in site 39 is the haematite content sufficiently high to show up as a definitive remanence carrier. A substantial amount of the haematite portion of site 39 is in the form of exsolved needles, bars or plates. The preferred alignment of these exsolved phases makes the magnetization of each grain quite anisotropic, but because of the random orientation of the individual grains the rock is inferred to be isotropic with respect to remanent magnetization.

The Curie balance curve (saturation magnetization versus temperature, J_s–T) of Fig. 5a is typical for the Askøy gabbro: a Curie temperature corresponding to that of nearly pure magnetite is generally found, and the cooling portion of the curve is invariably below the heating portion. The latter observation may be related to oxidation of magnetite, but due to the low specific magnetization of haematite its presence is normally completely obscured in the thermomagnetic analysis. Only in site 39 are there weak J_s–T indications for the presence of haematite (cf. Fig. 5b). Thus (1) the saturation magnetization (J_s) in this site is relatively weak suggesting a lower magnetite–haematite ratio than that normally encountered in the investigated sites (confirmed also by the microscope study), and (2) the 'paramagnetic plateau' of the J_s–T curves is reached c. 100 °C above that of other sites, i.e., above the Curie point of haematite.

In summary the oxide mineralogy precipitated during cooling of the Askøy gabbroic magma consisted primarily of relatively large grains of ferrianilmenite, titanhaematite and nearly pure magnetite. At the deuteric stage the first two phases underwent extensive exsolution processes and nearly pure ilmenite and haematite became cooling end products. Subsequent metamorphism has led to further purification of the ilmenite and haematite mineralogy (Fluge, 1986). In addition, partial chemical–mechanical conversion of large multidomain magnetite has resulted in mineral assemblages that provide magnetically more stable particles than those that formed during the original cooling.

5. The palaeomagnetic results in a geological framework

The combined geological and experimental evidence outlined in the foregoing section suggests that the Askøy pluton has experienced a complex magnetization history related to events of regional metamorphism. The radiometric information favours an original Precambrian age with subsequent metamorphism, but characteristic magnetizations of Precambrian relevance, for example, corresponding to poles 1–7 of Fig. 6 have not been found. It appears appropriate to conclude, therefore, that the Precambrian palaeomagnetic signatures have been completely erased by Caledonian overprinting.

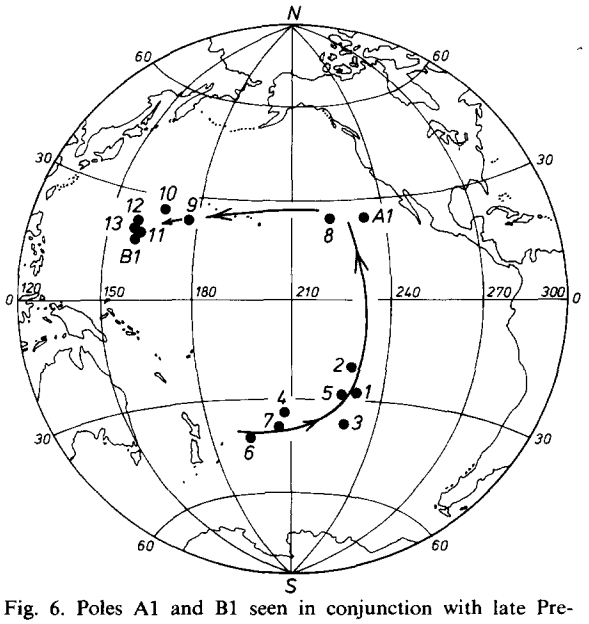


Fig. 6. Poles A1 and B1 seen in conjunction with late Precambrian (1–7) and Palaeozoic (8–13) poles for NW Europe. The reference poles are: 1, Egersun dolerites (Storetvedt and Gidskehaug, 1968); 2–4, Rogaland Precambrian (Porter, 1972); 5, Torridonian sediments (Stewart and Irving, 1974); 6, 7, anorthosites, S. Norway (Murthy and Deutsch, 1974, 1976); 8, pre-tectonic (ordovician) overprint of Dalradian tillites, Scotland (Stupavsky et al., 1982, slightly modified by Storetvedt and Deutsch, 1986); 9, Peterhead granite (Torsvik, 1985); 10, Foyers granite (Torsvik, 1984); 11, Kinghorn lavas (Wilson and Everitt, 1963); 12, Ringerike sandstone (Storetvedt et al., 1967); 13, Rørangen sediments (Storetvedt and Gjellestad, 1966).

Overall palaeomagnetic results for the two magnetization axes recorded, including statistics and pole positions, are summarized in Table II. As is seen in Fig. 6 the in situ poles, A1 and B1, accord extremely well with Palaeozoic poles from Scotland (east of the Great Glen Fault) and Norway. Thus the A1 pole is in close correspondence to the pre-tectonic overprinted pole of Dalradian tillites (Stupavsky et al., 1982; slightly modified by Storetvedt and Deutsch, 1986) inferred to be of Upper Ordovician age (Storetvedt and Deutsch, 1986). The younger pole, B1, fits mid-Palaeozoic data, agreeing particularly well with Devonian–Lower Carboniferous poles. We believe, therefore, that the ancient magnetization, now held by the pluton, is entirely of Caledonian age, having been imposed during Upper Ordovician and Devonian events of regional metamorphism. These suggested events of Caledonian remagnetization fit the available structural and radiometric evidence.

All the magnetite remanence has apparently been reset during the Devonian phase, while the Ordovician palaeomagnetic component, which seems to be carried entirely by haematite, has only survived sporadically. On this basis, it is not difficult to understand why the Precambrian palaeomagnetic component(s) is absent. In any case, judging from the opaque petrology, i.e., the inferred large multidomain particles of original magnetite, the investigated rocks would have most likely provided a very poor record of the original palaeofield. Some haematite precipitated at the time of cooling but large multidomain grains of nearly pure magnetite, presumably of low magnetic stability, originally formed the chief magnetic constituent. Post-cooling metamorphism has

TABLE II

Mean palaeomagnetic data for the mafic pluton of Askøy, West Norway

Group	N	D_m	I_m	R	k	α_{95}	Pole	δ_m	δ_p
A1	12	318	8	11	17	10	233E, 25N	10	5
A2	12	322	21				231E, 33N		
B1	35	025	−22	34	45	4	160E, 16N	3.7	2.0
B1 *	41	027	−20	39	22	5	158E, 16N	4.9	2.5
B2	35	016	−29				169E, 13N		

The A1 and B1 groups are based on data from Table I. Group B1* includes six specimen results that are of questional single-component nature (given a question mark in Table I). The A2 and B2 directions are according to a possible tectonic adjustment of the A1 and B1 results using primary fabric data. This correction implies a 20° westward rotation about a N–S horizontal axis. N: number of unit vectors (specimens); D_m: mean declination; I_m: mean inclination; α_{95}: half-angle of the cone of 95% confidence about the mean (in degrees); R: length of resultant vector; K: precision parameter; δ_p, δ_m: the semi-axes of the oval of 95% confidence about the mean pole. Statistics according to Fisher (1953).

led to further haematite formation and subdivision/disintegration of original magnetite, which provides a palaeomagnetically more stable basis for post-cooling magnetizations. However, in those parts of the pluton with the stronger textural alteration, the oxide mineralogy has to a major extent been converted to secondary iron-silicates leading to weak and inconsistent magnetizations.

Based on a certain asymmetry of primary fabric observations the possibility arises that the Askøy pluton may have been subjected to a westward tectonic rotation of c. 20° about a horizontal axis (Askvik 1971). However, Table II shows that the differences between uncorrected and corrected poles (A1 and B1 versus A2 and B2) are too small to justify any further discussion of this aspect, but the suggested tectonic rotation is certainly not ruled out by the present palaeomagnetic data. The very good agreement between the Askøy poles and other poles of Caledonian age for NW Europe suggests that the pluton has either been more or less in place since say Middle Ordovician or, if it (along with the Øygarden Gneiss Complex) was transported in post Middle Ordovician, the displacement did not involve motions including significant horizontal rotations.

The data presented here demonstrate the importance of comparing radiometric and palaeomagnetic results to establish a more complete age information. Thus, the preliminary radiometric dates are interpreted in terms of a Precambrian origin of the pluton with variable metamorphic resetting in Ordovician time. However, the question arises whether all ages obtained are composite, i.e., involve a Precambrian cooling age with partial resetting in post-Ordovician time. The fossil magnetism of the pluton has no longer any trace of its Precambrian origin but an Ordovician event of penetrative deformation is indeed confirmed. Furthermore, a younger Devonian age metamorphic event (probably of greenschist grade) is very well exposed by the palaeomagnetic data yet the radiometric results available so far from the Askøy pluton give no evidence for this younger phase of deformation.

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